Homework 4

(1) A standard deck of cards is shuffled well. Two cards are drawn randomly, one at a time without replacement. Let A be the event that the first card is a heart, and B be the event that the second card is red. Find $P(A \mid B)$ and $P(B \mid A)$.

(2) A coin is flipped twice. Assuming that all four outcomes in the sample space

$$S = \{(h, h), (h, t), (t, h), (t, t)\}$$

are equally likely, find the conditional probability that both flips land on heads, given that
  (a) the first flip lands on heads;
  (b) at least one flip lands on heads.

(3) In the card game bridge, the 52 cards are dealt out equally to 4 players, called East, West, North, and South. If North and South have a total of 8 spades among them, what is the probability that East has 3 of the remaining 5 spades?

(4) If $P(A) = 0.8$, $P(B) = 0.3$, and $P(A \cup B) = 0.86$, are the events A and B independent?

(5) In a certain college town, 25 percent of the students failed mathematics, 15 percent failed chemistry, and 10 percent failed both mathematics and chemistry. A student is selected at random.
  (a) If the student failed chemistry, what is the probability that he or she failed mathematics?
  (b) If the student failed mathematics, what is the probability that he or she failed chemistry?
  (c) What is the probability that the student failed mathematics or chemistry?
  (d) What is the probability that the student failed neither mathematics nor chemistry?

(6) Find $P(B \mid A)$ if:
  (a) $A \subseteq B$,
  (b) A and B are mutually exclusive (disjoint).
  Assume that $P(A) > 0$.

(7) Alana is undecided as to whether to take a French course or a chemistry course. She estimates that her probability of receiving an A grade would be $\frac{1}{2}$ in a French course and $\frac{2}{3}$ in a chemistry course. If Alana decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?

(8) Suppose that an urn contains 8 red balls and 4 white balls. Two balls are drawn from the urn without replacement. If we assume that at each draw every ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red?
  Provide two solutions:
  (a) Use the naive definition of probability.
  (b) Use the multiplication rule for probabilities.
  Make sure that the results coincide.

(9) An ordinary deck of 52 playing cards is randomly divided into two piles of 26 cards each. Compute the probability that each pile contains exactly 2 aces.

(10) There are three urns with the following compositions:
  Urn 1 contains 3 white balls and 2 black balls,
  Urn 2 contains 2 white balls and 5 black balls,
  Urn 3 contains 4 white balls and 3 black balls.
  One urn is selected by rolling a fair die as follows: the first urn is chosen if the outcome is 1, the second urn is chosen if the outcome is 2, and the third urn is chosen if the outcome is 3, 4, 5, or 6. After selecting the urn, one ball is drawn at random from it.
  Which is more likely: drawing a white ball or drawing a black ball?

(11) In a certain city, 40% of the people consider themselves Conservatives (C), 35% consider themselves Liberals (L), and 25% consider themselves Independents (I). During a particular election, 45% of the Conservatives voted, 40% of the Liberals voted, and 60% of the Independents voted. If a person is selected randomly, find the probability that he/she voted.